

St. Louis Pipe & Supply

CONFIDENTIAL



VALLOUREC STAR, LP

Seamless Tubular Products

Certified Test Report
11/28/2023

Order Line #: Q-221125736-1	Spec: SEAMLESS QUENCH AND TEMPERED	S LN (Heat): N92101
Grade: AP1SL X52Q 48th Edition May 1, 2019		W/O No: DE7158
OD (in): 10.750	Wall (in): 0.300	Lot: N638910
End Finish: PEB	Lbs/Ft: 54.79	Length: DRL
Comment: Melted and Manufactured in the USA Vallourec Star's QA program meets the req. of DIN EN 10204-2004 Type 3.1 Satisfies NACE MR0175 PAR3.2/ISO 15156-2: 2020 & ANSI/NACE MR0103/ISO 17945:2015 X52Q/ASTM A333-18 G1&G6 ASME SA333-2021 G1&G6 ASTM A106B/C-19a ASME SA106B/C-2021 ASTM A53B-22 ASME SA53B-2021 MEETS PSL2		

Mechanical Properties

Orientation	Type	Tensile Size Specimen Cross Section			Strength (ksi)			Elongation	
		Width (in)	Thick (in)	Area (sqin)	Yield	Tensile	Y/T Ratio	Gage Length	%Elong
LONGITUDINAL	STRIP	1.493	0.507	0.7597	65.0	77.8	0.84	2	44.0
LONGITUDINAL	STRIP	1.495	0.483	0.7247	66.5	78.7	0.85	2	43.0
LONGITUDINAL	STRIP	1.494	0.491	0.7362	65.7	77.8	0.84	2	44.0
Calculated Max Internal Yield (psi): 5294									

Supplemental Requirement

Austenitizing Temp	1700 °F
Tempering Temp	1315 °F
Flattening	PASSED
Hardness	SEE ATTACHED
Collapse Min	N/A
Collapse Actual	N/A
Hydrostatic Pressure	3,000 5 SEC HOLD
Other	N/A

Chemical Analysis

Chem Spec: 69Q

NDE SEE ATTACHED

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
HEAT	0.11	1.09	0.012	0.004	0.23	0.15	0.06	0.13	0.02	0.010	0.000	0.044	0.026	0.0014	0.0005	0.003	0.0096	0.34
PRODUCT 1	0.11	1.05	0.011	0.004	0.23	0.14	0.06	0.13	0.02	0.009	0.000	0.044	0.027	0.0016	0.0004	0.003	0.0085	0.34
PRODUCT 2	0.11	1.05	0.011	0.004	0.23	0.15	0.06	0.13	0.02	0.009	0.000	0.044	0.027	0.0016	0.0004	0.003	0.0086	0.33

Charpy Impact Testing

	Notch	Direction	Size	Temp °F	Position	Ft Lbf 1	Ft Lbf 2	Ft Lbf 3	Ft Lbf Avg	Shear 1	Shear 2	Shear 3	Shear Avg
Min Target	—	TRANS	FULL	32	—	15	15	15	20				
Actual	V	TRANS	FULL	32	TRAIL	320	325	322	322	100	100	100	100
Min Target	—	TRANS	FULL	-50	—	10	10	10	13				
Actual	V	TRANS	FULL	-50	TRAIL	241	266	253	253	100	100	100	100
Actual	V			-50	TRAIL	241	266	253	253	100	100	100	100

NDT Inspection: WMSEA

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star, without the expressed written consent of Vallourec Star negates the validity of the test report. Vallourec Star is not responsible for the inability of the material to meet specific applications.

Signed: 
 Name: PETER YANG
 Date Approved: 11/28/2023

DRAFT

 Production Bend Report			
Customer:	DNOW	Report No.:	105093-PBR-1
Customer PO No.:	662682056	Code of Fabrication:	
TTB Job No.:	25H105093	ASME B16.49-2023	

Mother Pipe & Bend Information

OD (in): 10.750 Grade: API-5L-X52 PSL 2 Radius (in): 50 Bending Machine ID: M12 Heat No.: N90101

Essential Variables

Wall Thickness (inch): 0.500 Radius-to-Diameter Ratio: 5.0 Coil Design: TTB High Spec Coolant Type: Water

Weld Seam Location: n/a Heat Treatment Holding Temp (°F): 1050 Heat Treatment Soak Time (min): 60.0

Note 1: Bending recipe (Bend Speed & Bend Temp parameters) are proprietary to Tulsa Tube Bending Co. and can be made available to the purchaser or the purchaser's third party inspector(s) at the bend facility.

Bend ID No.	Operator Initials	Date	Bending Speed (ipm)	Extrados Bend Temp (°F)	Intrados Bend Temp (°F)	Coolant Temp (°F)	Coolant Flow Rate (GPM)	Induction Frequency (Hz)
Qualification Bend Value:			See Note 1.			72.3	10.0	871
Limits of Variation:			+/- 0.1 ipm	+/- 50°F	+/- 50°F	+/- 25°F	+/- 10%	+/- 20%
105093-1-1	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	69.0	10.0	880
105093-1-2	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	70.0	10.0	881
105093-1-3	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	70.0	10.0	880
105093-1-4	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	70.0	10.0	880
105093-1-5	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	71.0	10.0	882
105093-1-6	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	72.0	10.0	881
105093-1-7	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	71.0	10.0	882
105093-1-8	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	72.0	10.0	880
105093-2-1	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	71.0	10.0	881
105093-2-2	MH	* 4/14/2025	+/- 0.1 ipm	+/- 50°F	+/- 50°F	72.0	10.0	880

* Revised 5/22/25 to reflect actual date of production.

 5/22/25

Customer Information

Your PO 25H105093-1

105093-1-1

105093-1-2

105093-1-3

105093-1-4

105093-1-5

105093-1-6

105093-1-7

105093-1-8

105093-2-1

105093-2-2

Thermal Specialties

Heat Treating Certification

Thermal Specialties. Information

Date Processed: 04/17/2025

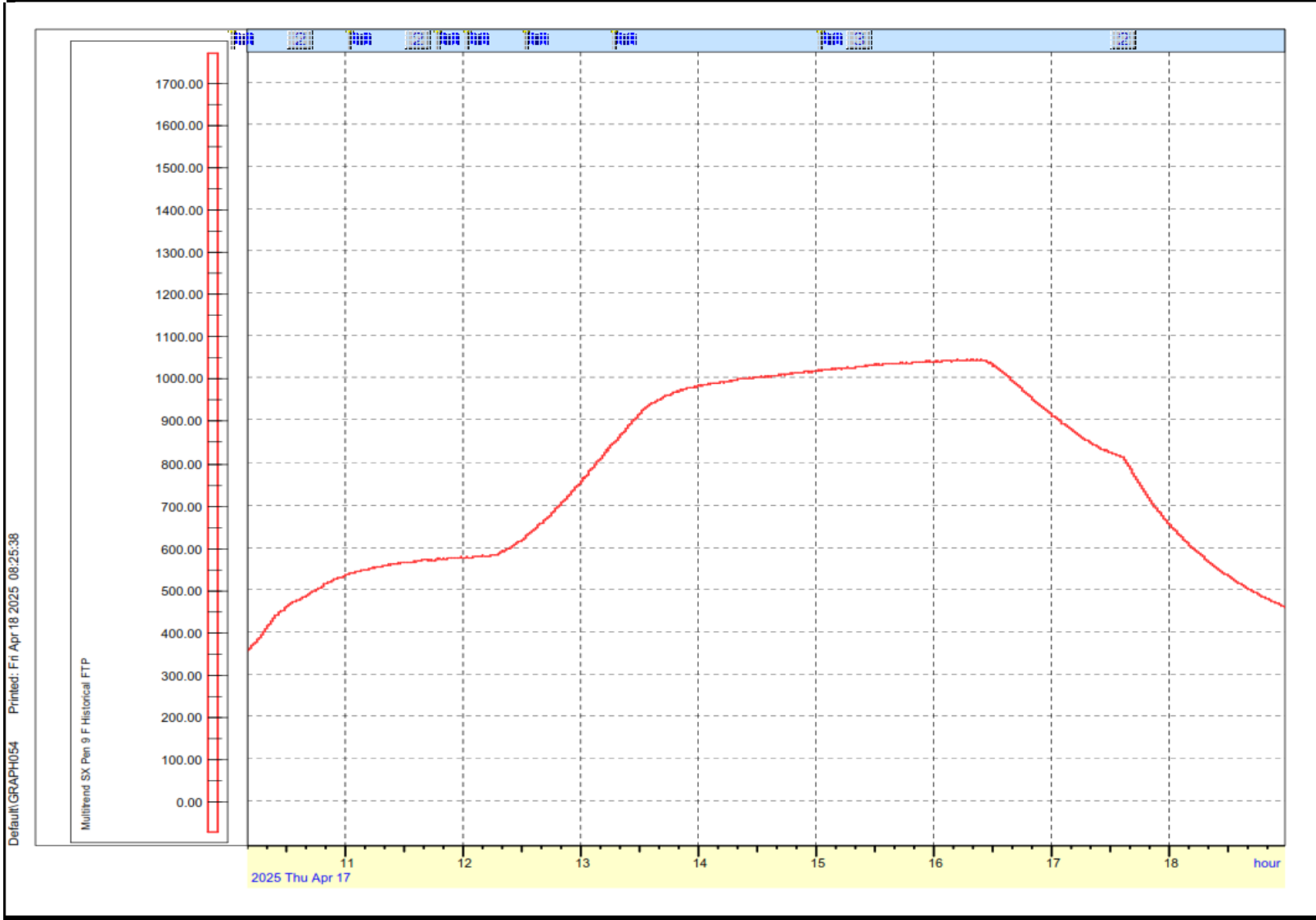
Order ID: 299356 tc load

Furnace#: 27

Calibration Date: 11/19/2024

Thermocouple Type K

Approved by: JM



Thermal Specialties certifies that the item has been processed in accordance with customer's requirements.
Processes were obtained through standard approved methods and in accordance with Thermal Specialties quality assurance manual



Production Bend Hardness Report

Customer:	DNOW	Report No.:	105093-H-1
Customer PO No.:	662682056	Code of Fabrication: ASME B16.49-2023	
TTB Job No.:	25H105093		

Mother Pipe Information

Process Information

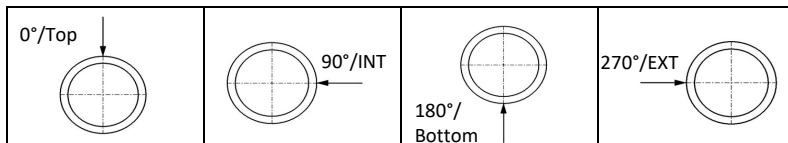
Pipe Size (in): 10.75 x 0.500

Grade: API-5L-X52 PSL 2

Instrument Model No.: GE MIC-10

Serial No.: 1-1718

Key:



T= Top (Weld if Welded Material)

H= Weld Heat Affected Zone

I= Intrados of Bend

E= Extrados of Bend

TAN= Tangent of Bend

STZ= Start Transition Zone

FTZ= Finish Transition Zone

3/4= 3/4 Point of the Bend Arc

Note 1. Weld & Weld HAZ readings are only required on bends made from welded mother pipe.

Note 2. Readings at 3/4 bend arc locations are only required for production bends at least 30° greater than the QB angle.

Hardness Readings

<u>Hardness Readings</u>		Min Allowed Values:				137		Max Allowed Values:			238		3/4= 3/4 Point of the Bend Arc			
	TAN T	TAN H	TAN	STZ E	STZ I	MID T	MID H	MID E	MID I	FTZ E	FTZ I	3/4 T	3/4 H	3/4 EXT	3/4 INT	
Bend ID No.	0°	0°	270°	270°	90°	0°	0°	270°	90°	270°	90°	0°	0°	270°	90°	
105093-1-1	N/A	N/A	190	180	180	174	N/A	179	176	176	186	N/A	N/A	N/A	N/A	
105093-1-2	N/A	N/A	189	179	175	180	N/A	179	169	170	170	N/A	N/A	N/A	N/A	
105093-1-3	N/A	N/A	191	170	179	176	N/A	179	176	171	183	N/A	N/A	N/A	N/A	
105093-1-4	N/A	N/A	178	169	170	170	N/A	176	168	173	171	N/A	N/A	N/A	N/A	
105093-1-5	N/A	N/A	185	173	177	177	N/A	178	175	172	180	N/A	N/A	N/A	N/A	
105093-1-6	N/A	N/A	178	180	178	174	N/A	181	181	177	175	N/A	N/A	N/A	N/A	
105093-1-7	N/A	N/A	185	179	172	185	N/A	173	180	173	174	N/A	N/A	N/A	N/A	
105093-1-8	N/A	N/A	179	179	175	184	N/A	180	178	180	173	N/A	N/A	N/A	N/A	
105093-2-1	N/A	N/A	175	175	171	178	N/A	180	175	173	175	N/A	N/A	N/A	N/A	
105093-2-2	N/A	N/A	177	180	189	178	N/A	179	182	177	179	N/A	N/A	N/A	N/A	

READINGS:



MEET ACCEPTANCE CRITERIA

PERFORMED BY:

JV

DATE:

4/22/2025



DO NOT MEET ACCEPTANCE CRITERIA



Dimensional-Visual Inspection Report

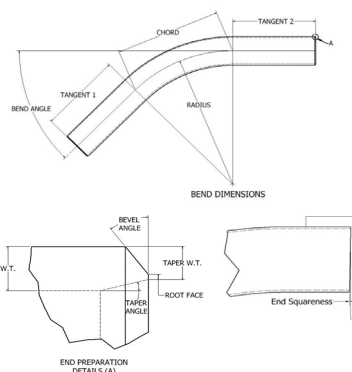
Customer:	DNOW	Pipe Size (in):	10.75 x 0.500	Report No.:	105093-DV-1-3
Customer PO No.:	662682056	Grade:	API 5L X52 PSL 2	Bend ID No.:	105093-1-3
TTB Job No.:	25H105093	Heat No.:	N90101	Code of Fabrication:	ASME B16.49-2023

Physical Dimensions

Dimension	Target	Tolerance	Actual
Tangent 1 (in)	24 MIN	MIN	26.13
Tangent 2 (in)	24 MIN	MIN	25.88
Chord (in)	25.9	n/a	26.3
Angle (deg)	30.0	+/- 0.5 °	30.5
Radius, R (in)	50	+/- 1%	49.9
Performed By:	BB	Date:	4/24/2025

End Prep Dimensions

Dimension	Target	Tolerance	Actual	
Bevel Angle (deg)	37.5	+/- 2.5°	37.5	END 1
Root Face (in)	0.06	+/- 0.03 in	0.060	
Taper Angle (deg)	n/a	n/a	n/a	
Min ID	9.75	+/- 0.1 in	9.750	
Max ID	9.75	+/- 0.1 in	9.800	
Max End Sq. (in)	0	+/- 0.09 in	0.000	END 2
Bevel Angle (deg)	37.5	+/- 2.5°	37.500	
Root Face (in)	0.06	+/- 0.03 in	0.030	
Taper Angle (deg)	n/a	n/a	n/a	
Min ID	9.75	+/- 0.1 in	9.750	
Max ID	9.75	+/- 0.1 in	9.800	
Max End Sq. (in)	0	+/- 0.09 in	0.030	
Performed By:	BB	Date:	4/24/2025	



Ovality [(O.D. max - O.D. min)/O.D. nominal] x 100

Quadrant	Limit	Min OD	Max OD	% Ovality
End 1	1.0%	10.729	10.758	0.3%
Start of Bend	3.0%	10.644	10.788	1.3%
1/4 of Bend Arc	3.0%	10.672	10.763	0.8%
1/2 of Bend Arc	3.0%	10.672	10.774	0.9%
3/4 of Bend Arc	3.0%	10.685	10.740	0.5%
Stop of Bend	3.0%	10.696	10.735	0.4%
End 2	1.0%	10.749	10.759	0.1%
Performed By:	BB	Date:	4/24/2025	

Wall Thinning of Extrados

Quadrant	Limit	Nominal Wall Thickness	Extrados Wall Thickness	% Wall Thinning
Start of Bend	10.0%	0.500	0.468	6.4%
1/4 of Bend Arc	10.0%	0.500	0.462	7.6%
1/2 of Bend Arc	10.0%	0.500	0.460	8.0%
3/4 of Bend Arc	10.0%	0.500	0.466	6.8%
Stop of Bend	10.0%	0.500	0.468	6.4%
Performed By:	BB	Date:	4/24/2025	

ID Gauge Drift/Pig Inspection

Minimum Gauge Diameter =	97.0%	of Nominal ID, or	9.458 in
Gauge Length Required =	9.75 in		
Gauge passes through the entire bend without power equipment assistance			
Performed By:	BB	Date:	4/24/2025

Visual Inspection

All accessible surfaces found to be free of visible injurious defects.	
Bend is stenciled with correct information.	
Performed By:	BB
Date:	4/24/2025

READINGS: ☒ MEET ACCEPTANCE CRITERIA
☐ DO NOT MEET ACCEPTANCE CRITERIA

GLOBE X-RAY ASSUMES NO RESPONSIBILITY FOR THE INTEGRITY OF THE STRUCTURE OR FOR ANY LOSSES OF ANY KIND DUE TO ANY KIND OF INTERPRETATION.



Tulsa Tube Bending
4192 S. Galveston Ave.
Tulsa, OK 74107
Tel: 918-446-4461
Fax: 918-446-1595

Serving others. Building people. Pursuing excellence.

Certificate of Conformance

Customer: DNOW Certificate No.: 20250424

Customer PO No.: 662682056 Issue Date: 4/24/2025

TTB Job No.: 25H105093

Material Size, Grade: 10.75 inch OD x 0.500 inch w.t., API-5L-X52 PSL 2 SMLS Quantity of Bends: 10

Material Heat No(s): N90101

Bend ID Nos.:	<u>105093-1-1</u>	<u>105093-1-2</u>	<u>105093-1-3</u>	<u>105093-1-4</u>	<u>105093-1-5</u>
	<u>105093-1-6</u>	<u>105093-1-7</u>	<u>105093-1-8</u>	<u>105093-2-1</u>	<u>105093-2-2</u>

We hereby certify that the aforementioned induction bends have been inspected and conform to all of the requirements of the Purchase Order. The induction bends have been manufactured in conformance with:

ASME B16.49-2023

*With the following notes and exceptions:

- Maximum wall thinning shall not exceed 10% from nominal.

- Bends to be beveled per ASME 37.5° (±2.5°) per B16.49/B16.25.

Approved By: Austin Carter Date: 4/24/2025

FBE Coating per customer coating spec Doc ID TPE_26907sp.

*With the following notes and exceptions:

- Section 9.0 – Laboratory testing not included.

- Section 6.1 – Due to the hand applied nature of the application, max DFT may exceed 36 mils in isolated areas.

- Section 6.2.9 – Acid wash is not available.

- Section 6.3.2 – Heat indicating Temp Stick may be used but will be applied near the cutback area with use kept to a minimum.



COATING INSPECTION RECORD

KF-010 FBE 5/27/22 REV2

CUSTOMER: TULSA TUBE BENDING	DATE: 4/29/2025	ITEMS: 5 PCS
CUSTOMER PO: 25H105093	PAINT SPEC: CONSUMERS ENERGY COMPANY	

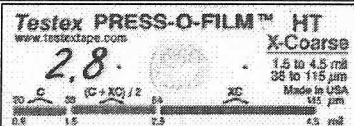
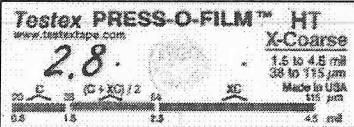
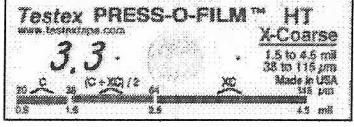
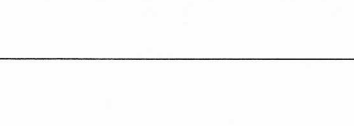
SURFACE PREPARATION

ITEM SALT TEST	N/A ☐	RESULT: 0	SSPC SP-1 IF NEEDED	DATE	4/29/2025	4/29/2025
ABRASIVE SALT TEST	N/A ☐	RESULT: 1	CLEANLINESS REQUIRED	TIME	10:00AM	12:00PM
PH OF ABRASIVE	N/A ☒	RESULT:	CLEANLINESS ACHIEVED	AIR TEMP	81°F	80°F
TYPE OF ABRASIVE	STEEL	PROFILE REQUIRED	2-4 MILS	RELATIVE HUMIDITY	65%	69%
NOZZLE PSI	100PSI	PROFILE ACHIEVED	☒ YES ☐ NO	STEEL	85°F	87°F
BLOTTER TEST	☒ PASS	SSPC-VIS 1 USED	☒ YES ☐ NO	DEW POINT	68°F	69°F
DUST FREE SURFACES	☒ YES ☐ NO	VISUAL INSPECTION	☒ YES ☐ NO	STAGE OF OPERATION	PRE-BLAST	PRE-COATING

COATING APPLICATION

COAT 1 ST	PRODUCT NAME: VALSPAR 2000	BATCH: CV3444JP	TARGET MILS: 16	TARGET PRE COAT TEMP: 450 DEG F
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FINAL TESTING

HVST @ 2000 VOLTS PASS ☒		ADHESION KNIFE TEST: PASS ☒		DFT TEST: PASS ☒		ALL REQUIREMENTS MET: PASS ☒	
KI#	DESCRIPTION	MIL AVERAGE	PRESS O FILMS	KI#	DESCRIPTION	MIL AVERAGE	
1-3	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	28					
1-6	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	24					
1-7	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	26					
2-1	10.75" X .500" WT API5L X52 SMLS 45 DEG BEND	25					
2-2	10.75" X .500" WT API5L X52 SMLS 45 DEG BEND	23					

VERIFIED BY KEITH:

[Signature]



COATING INSPECTION RECORD

KF-010 FBE 5/27/22 REV2

CUSTOMER: TULSA TUBE BENDING	DATE: 4/30/2025	ITEMS: 5 PCS
CUSTOMER PO: 25H105093	PAINT SPEC: CONSUMERS ENERGY COMPANY	

SURFACE PREPARATION

ITEM SALT TEST	N/A ☐	RESULT: 0	SSPC SP-1 IF NEEDED	DATE	4/30/2025	4/30/2025
ABRASIVE SALT TEST	N/A ☐	RESULT: 1	CLEANLINESS REQUIRED	TIME	10:00AM	12:00PM
PH OF ABRASIVE	N/A ☒	RESULT:	CLEANLINESS ACHIEVED	AIR TEMP	80°F	82°F
TYPE OF ABRASIVE	STEEL	PROFILE REQUIRED	2-4 MILS	RELATIVE HUMIDITY	71%	67%
NOZZLE PSI	100PSI	PROFILE ACHIEVED	☒ YES ☐ NO	STEEL	83°F	88°F
BLOTTER TEST	☒ PASS	SSPC-VIS 1 USED	☒ YES ☐ NO	DEW POINT	70°F	70°F
DUST FREE SURFACES	☒ YES ☐ NO	VISUAL INSPECTION	☒ YES ☐ NO	STAGE OF OPERATION	PRE-BLAST	PRE-COATING

COATING APPLICATION

COAT 1 ST	PRODUCT NAME: VALSPAR 2000	BATCH: CV3444JP	TARGET MILS: 16	TARGET PRE COAT TEMP: 450 DEG F
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FINAL TESTING

HVST @ 2000 Volts PASS ☒			ADHESION KNIFE TEST: PASS ☒		DFT TEST: PASS ☒		ALL REQUIREMENTS MET: PASS ☒	
KI#	DESCRIPTION	MIL AVERAGE	PRESS O FILMS			KI#	DESCRIPTION	MIL AVERAGE
1-1	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	26						
1-2	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	23						
1-4	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	24						
1-5	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	20						
1-8	10.75" X .500" WT API5L X52 SMLS 30 DEG BEND	19						

VERIFIED BY KEITH:

John Keith